

AI Impact on Jobs 2030

Semester Project (Spring 2026)

Muhammad Haseeb UI Hassan

Course: STAT 206 – Introduction to Data Analytics, Section A

Instructor: Dr. Shakila Bashir

Institution: Forman Christian College (A Chartered University)

Term: Spring 2026 | Dataset: AI Impact on Jobs 2030 (Kaggle, n = 100)

Introduction & Research Objectives

BACKGROUND

Artificial Intelligence is transforming global labor markets at an unprecedented pace. As automation probability rises across sectors, understanding how salary, experience, education, and AI exposure interrelate is critical for workforce planning.

This project applies inferential statistics to a 100-observation Kaggle dataset — *AI Impact on Jobs 2030* — to test whether traditional predictors of salary remain valid in an AI-disrupted job market.

RESEARCH OBJECTIVES

1. Summarize salary, experience, AI exposure, and automation probability using descriptive statistics
2. Visualize distributions and relationships across risk and education groups
3. Test whether salary differs by experience level (t-test) and education (ANOVA)
4. Examine independence of education and automation risk (chi-square)
5. Quantify correlations between salary and key predictors
6. Build a simple linear regression of salary on years of experience

Dataset Overview

Source: Kaggle – AI Impact on Jobs 2030 | **Observations:** n = 100 | **Analysis Tool:** R (ggplot2)

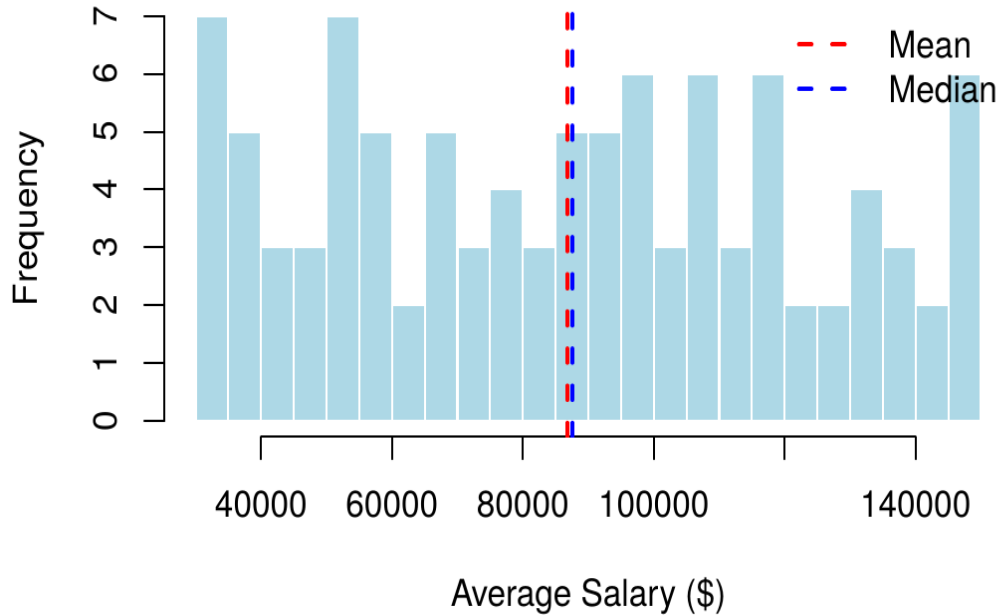
Variable	Type	Description	Role in Analysis
Average Salary (\$)	Continuous	Annual salary in USD	Dependent variable (Y)
Years of Experience	Continuous	Total work experience in years (0–29)	Predictor / grouping variable
AI Exposure Index	Continuous (0–1)	Degree of job exposure to AI technologies	Correlation predictor
Automation Probability 2030	Continuous (0–1)	Estimated probability of automation by 2030	Correlation predictor
Risk Category	Categorical (3 levels)	Low / Medium / High automation risk	Grouping variable
Education Level	Categorical (4 levels)	High School / Bachelor's / Master's / PhD	Grouping variable (ANOVA, Chi-sq)

Descriptive Statistics

Statistic	Average Salary (\$)	Years Experience	AI Exposure Index	Automation	KEY INSIGHTS
Mean	\$86,807	15.19 yrs	0.510	0.500	Salary Mean ≈ Median → near-symmetric distribution. CV = 40.3% signals wide wage dispersion. Experience Mean 15.2 yrs, CV = 58.9% → highly variable sample. AI & Automation Both near 0.50 mean — dataset spans full range of AI exposure levels.
Median	\$87,576	15.0 yrs	0.530	0.520	
Std. Dev.	\$34,989	8.95 yrs	0.300	0.260	
Variance	1,224,257,557	80.07	0.0902	0.0697	
Min	\$30,060	0 yrs	0.01	0.05	
Max	\$148,721	29 yrs	0.99	0.95	
Range	\$118,661	29 yrs	0.98	0.90	
Q1	\$55,384	7.75 yrs	0.218	0.288	
Q3	\$114,552	23.25 yrs	0.755	0.705	
IQR	\$59,168	15.5 yrs	0.537	0.417	
CV (%)	40.31%	58.91%	58.59%	52.30%	

Graph 1: Histogram of Average Salary

Distribution of Average Salary



INTERPRETATION

Distribution Shape

Near-uniform salary distribution spanning \$30K–\$149K. Mean (\$86,807) $\approx$ Median (\$87,576), confirming approximate symmetry with no strong skew.

Dispersion

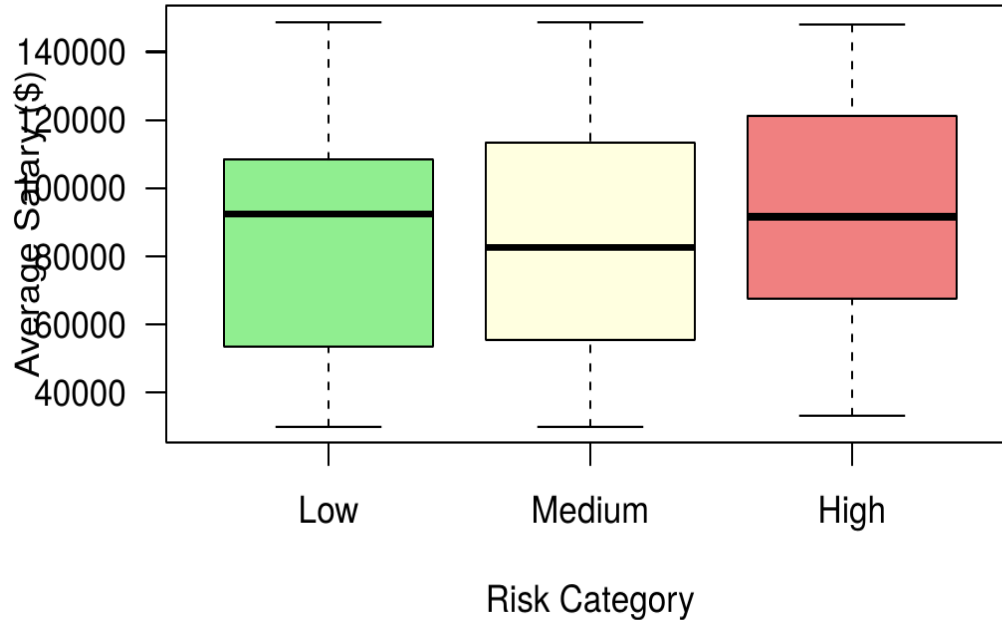
CV = 40.31% and SD = \$34,989 indicate substantial wage dispersion, reflecting diverse job types and industries.

Implication

Wide spread suggests salary is influenced by many unmeasured factors beyond those in this dataset.

Graph 2: Salary Distribution by Risk Category

Average Salary by Risk Category



INTERPRETATION

Overlapping IQRs

All three categories (Low, Medium, High) show very similar median salaries and overlapping IQRs — automation risk alone does not differentiate compensation.

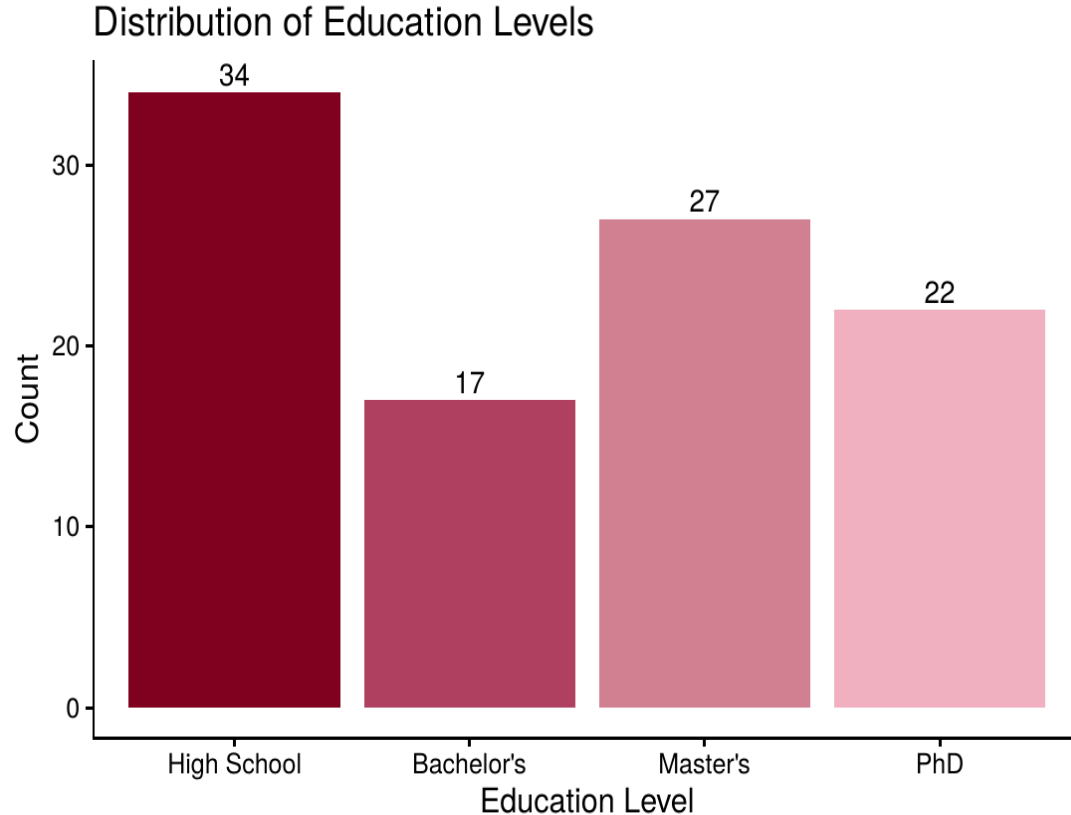
Visual Support for ANOVA

This boxplot visually supports the ANOVA finding ($F = 0.164$, $p = 0.920$) — no significant salary difference across risk groups.

Takeaway

Being in a high-risk automation job does not currently correlate with lower pay in this sample.

Graph 3: Distribution of Education Levels



INTERPRETATION

Unequal Distribution

High School dominates (34%), followed by Master's (27%), PhD (22%), and Bachelor's (17%) — groups are unequally sized.

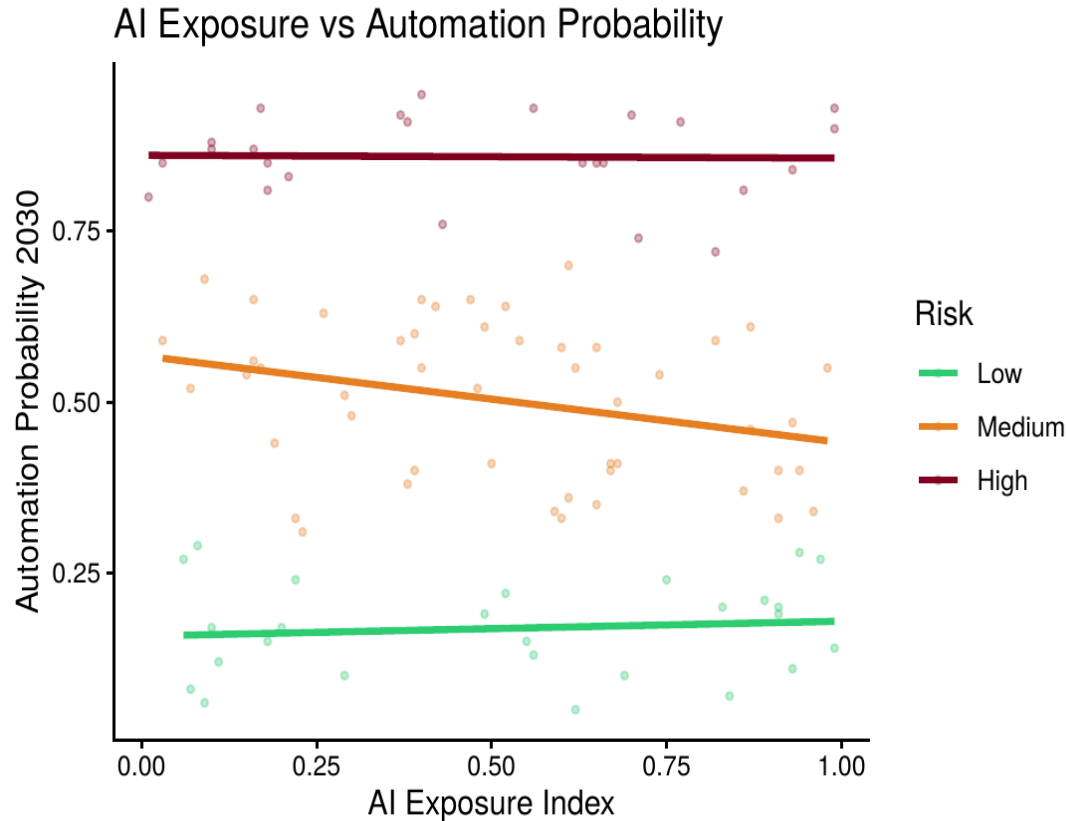
Statistical Power

Unequal group sizes reduce statistical power for detecting salary differences in the ANOVA test.

Notable Pattern

The prevalence of HS-educated workers (34%) in an AI jobs dataset suggests entry-level roles are well-represented alongside highly educated workers.

Graph 4: AI Exposure vs Automation Probability by Risk



INTERPRETATION

Risk Category Separation

Risk groups separate clearly along the Automation Probability axis — Low jobs cluster low, High jobs cluster high — confirming Risk_Category is derived from Automation_Probability_2030.

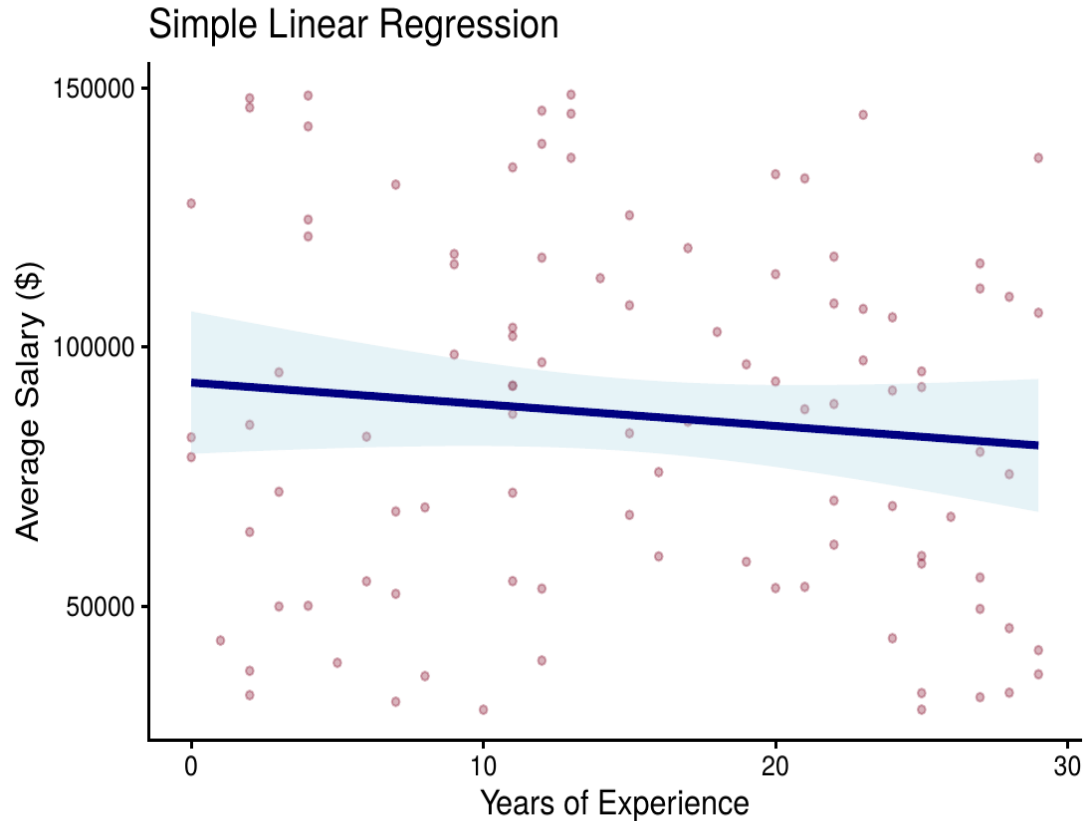
AI Exposure Independence

AI Exposure Index shows no systematic pattern within any risk group — it measures a different construct than automation probability.

Implication

Exposure to AI tools $\neq$ being automated. A job may use AI heavily yet face low automation risk.

Graph 5: Years of Experience vs Average Salary



INTERPRETATION

Nearly Flat Regression Line

Slope = $-\$417.94$ is trivially small. The wide confidence band reflects high uncertainty — points scatter randomly with no discernible pattern.

Negligible R^2

$R^2 = 0.0114$ — experience explains only 1.14% of salary variance. The model overall is not significant ($p = 0.290$).

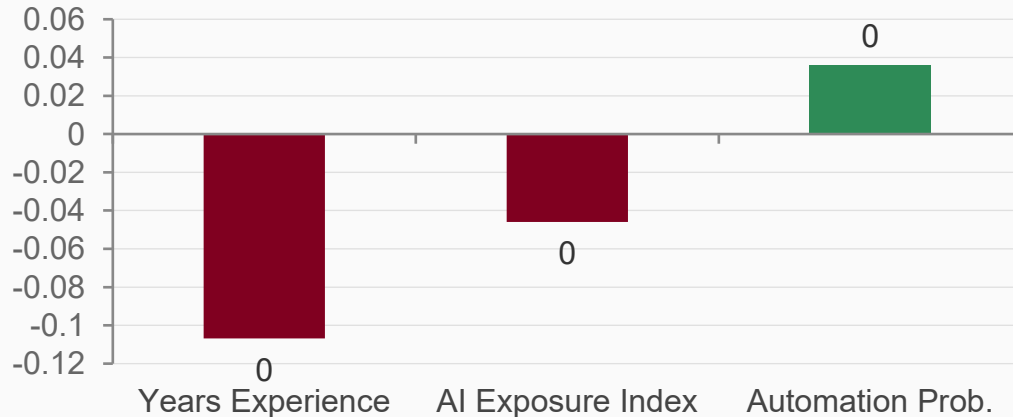
Counterintuitive Finding

In traditional markets, experience drives pay. Here, salary appears decoupled from experience — consistent with an AI-disrupted job market.

Correlation Analysis

Predictor	Pearson r	p-value	Spearman r	p-value	Kendall τ	Sig.
Years Experience	-0.1069	0.2898	-0.0917	0.3640	-0.0697	X
AI Exposure Index	-0.0459	0.6499	-0.0498	0.6224	-0.0276	X
Automation Prob.	+0.0357	0.7241	+0.0272	0.7885	+0.0193	X

Pearson r vs Average Salary



INTERPRETATION

All $|r| < 0.11$ — negligible correlations. All p-values exceed 0.05 (α). Pearson, Spearman, and Kendall results all agree — findings are not driven by outliers.

The weak negative r with experience (-0.107) is counterintuitive but consistent — salary appears randomly distributed regardless of experience in this dataset.

Hypothesis Testing

Test 1: Independent Samples t-Test

Salary: Low Experience (≤ 10 yrs) vs High Experience (> 10 yrs)

$H_0: \mu_{\text{low}} = \mu_{\text{high}}$ (no salary difference by experience group) | $H_1: \mu_{\text{low}} \neq \mu_{\text{high}}$

Low: $n=31$, \$83,209 (SD \$39,490) | High: $n=69$, \$88,423 (SD \$32,952)

$t = 0.6875$, $p = 0.2467 > 0.05 \rightarrow$ **FAIL TO REJECT H_0** | The \$5,215 difference is not statistically significant. Could be random sampling variation.

Test 2: One-Way ANOVA (F-Test)

Salary across Education Levels (HS / Bachelor's / Master's / PhD)

$H_0: \mu_{\text{HS}} = \mu_{\text{Bach}} = \mu_{\text{Mast}} = \mu_{\text{PhD}}$ | H_1 : At least one group mean differs

HS: \$88,063 | Bach: \$87,288 | Mast: \$82,860 | PhD: \$89,338

$F = 0.1643$, $p = 0.9202 >> 0.05 \rightarrow$ **FAIL TO REJECT H_0** | Group means differ by only \$6,478. Education level has no significant effect on salary.

Test 3: Chi-Square Test of Independence

Education Level vs Risk Category

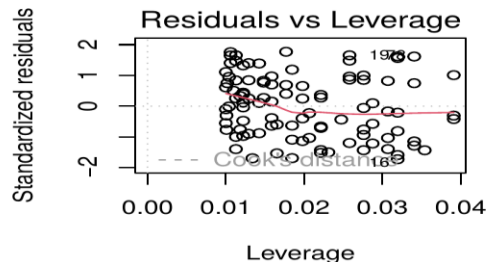
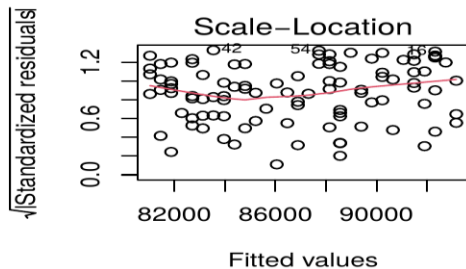
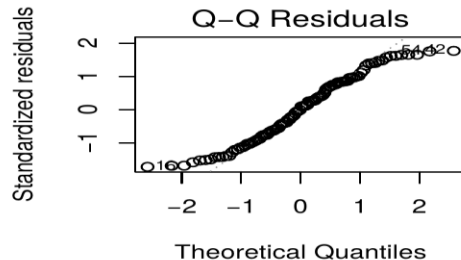
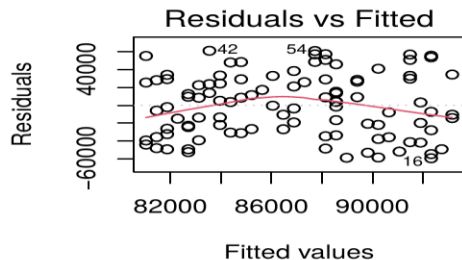
H_0 : Education level and automation risk are independent | H_1 : They are associated

Observed vs Expected frequencies closely matched across 4×3 table

$\chi^2 = 2.7271$, $df = 6$, $p = 0.8422 >> 0.05 \rightarrow$ **FAIL TO REJECT H_0** | Education and risk category are statistically independent. Higher education does not protect against automation risk.

Simple Linear Regression: Salary ~ Years of Experience

$$\text{Salary} = 93,155.33 - 417.94 \times \text{Years_Experience}$$



MODEL RESULTS & DIAGNOSTICS

Intercept $\beta_0 = \$93,155.33$

$p < 0.001$ — significant baseline salary

Slope $\beta_1 = -\$417.94 / \text{year}$

$p = 0.290$ — NOT significant | $SE = \$392.72$

$R^2 = 0.0114$ (1.14% variance explained)

98.86% of salary variation remains unexplained

Residual $SE = \$34,966$ | $F p = 0.290$

Diagnostics: Residuals random ✓ Q-Q near diagonal ✓
No influential outliers ✓ Low R^2 ✗

Conclusion & Recommendations

01 No Significant Salary Predictors

Experience, education, AI exposure, and automation probability all failed to significantly predict salary. All correlations $|r| < 0.11$, all p-values > 0.05 .

02 Education ≠ Higher Salary

ANOVA ($F = 0.164$, $p = 0.920$): salary means across HS, Bachelor's, Master's, and PhD differ by only \$6,478 — statistically negligible.

03 Automation Risk ≠ Lower Pay

Boxplot and chi-square ($\chi^2 = 2.73$, $p = 0.842$) confirm risk category and education are independent — being high-risk does not reduce compensation.

04 Regression Explains Only 1.14%

SLR model is non-significant ($F p = 0.290$). Salary in the AI-era appears driven by unobserved factors: specific skills, industry, geography, company size.

Recommendation: Future research should include sector, location, and skills data to identify true salary drivers in the AI labor market.

Thank You

Muhammad Haseeb Ul Hassan

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Dataset: AI Impact on Jobs 2030 — Kaggle | Analysis: R (ggplot2)